

MARCH, 2026

REACH FOR CHANGE

PART 1: UNCOVERING DONOR PROFILES THROUGH CLUSTERING

DATA SCIENCE AND MACHINE LEARNING
GROUP PROJECT 2025/2026



I. INTRODUCTION

In recent years, nonprofit organizations have faced an unexpected challenge.

While the number of charitable causes has grown significantly, people's tolerance for repeated solicitations has decreased. Many potential supporters report feeling overwhelmed by generic fundraising campaigns, leading to lower engagement and long-term disengagement from charitable initiatives.

To address this issue, the **Civic Support Alliance (CSA)**, a federation representing multiple humanitarian and social aid programs, has decided to modernize how outreach campaigns are conducted.

Instead of contacting everyone in their database, the organization wants to better understand the different types of individuals within their supporter base. The objective is not only operational efficiency but also respect: communicating in ways that are more relevant, more personalized, and better aligned with people's profiles and past behavior.

The CSA has accumulated historical data from previous fundraising campaigns, including demographic information, past interaction history, and donation behavior. They have provided a dataset containing individuals who may be contacted in an upcoming campaign. However, before building predictive tools, the organization first needs **a clearer picture of who these individuals are**.

Your team has been hired as a group of data scientists to conduct an exploratory segmentation study. Your goal is to use **clustering techniques to identify meaningful groups of individuals within the dataset**.

By uncovering distinct supporter profiles, you will help the organization understand the structure of its donor base and lay the foundation for more informed and targeted outreach strategies.

II. PROJECT GOALS

The primary components and goals of the project are:

- 1. Preprocessing and Exploratory Data Analysis (EDA):** Clean and preprocess your dataset, performing an exploratory analysis to uncover patterns and relationships.
- 2. Clustering:** Identify and analyze reasonable segments with, at least, two perspectives. To do that, you will need to choose different features for each segment, implement clustering models, tune them, evaluate the output, **report all your findings**, choose the one it makes more sense to you, **and provide a critical assessment**. Your focus is on segmenting the population, analyzing the key features that define each group and understanding what differentiates them. This should be accompanied by **a preliminary action plan** that will serve as a guide for future initiatives.

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III. DATASET

You have access to one datasets:

donors_train.csv: Containing features that give detailed information about each possible donor.

The available data contains the following attributes:

ATTRIBUTE	DESCRIPTION
CONTROL_NUMBER	Unique identifier of each possible donor
DONOR_AGE	Age as of last year's mail solicitation
URBANICITY	Nature of the donor's location U = Urban C = City T = Town R = Rural S=Suburban
SES	Code assigning each individual to one of 5 demographic profiles based on socioeconomic characteristics
HOME_OWNER	H if individual is a Homeowner, U if it is Unknown
DONOR_GENDER	Gender of individual
INCOME_GROUP	Code assigning each individual to one of 7 income group levels
WEALTH_RATING	Code assigning each individual to one of 10 wealth rating groups
MEDIAN_HOME_VALUE	Individuals' median home value in \$100's
MEDIAN_HOUSEHOLD_INCOME	Individuals' median household income in \$100's
PCT_OWNER_OCCUPIED	Percent of owner-occupied housing near where person lives
PER_CAPITA_INCOME	Per Capita Income of neighborhood in which person lives
PCT_ATTRIBUTE1	Percent of residents in person's neighborhood that are male and active military
PCT_ATTRIBUTE2	Percent of residents in person's neighborhood that are male and veterans
PCT_ATTRIBUTE3	Percent of residents in person's neighborhood that are Vietnam veterans
PCT_ATTRIBUTE4	Percent of residents in person's neighborhood that are WW2 veterans
PEP_STAR	If person achieved STAR donor status, given to individuals who have given to 3 consecutive campaigns.
RECENT_STAR_STATUS	If achieved STAR within last 4 years.
CHILDREN	Number of children the individual has
RECENT_AVG_GIFT_AMT	Average donation from the individual to the charitable organization in the last four years.
RECENT_CARD_RESPONSE_PROP	Number of times the individual has responded to a card solicitation from the charitable organization in the last four years

ATTRIBUTE	DESCRIPTION
REGENCY_STATUS_96NK	Donor status: A (active) - made their first donation more than 12 months ago and has made a donation in the last 12 months.); E (inactive) - made a donation 25+ months ago.); F (first time) - made their first donation in the last 6 months and has made just one donation.); L (lapsing) - made their last donation between 13-24 months ago); N (new) - made their first donation in the last 12 months and is not a First time donor); S (star donor)
FREQUENCY_STATUS_97NK	Number of times donated in last period (The period of recency for all groups except L and E is the last 12 months. For L it is 13-24 months ago, and for E it is 25-36 months ago).
RECENT_RESPONSE_PROP	Proportion of responses to the individual to the number of (card or other) solicitations from the charitable organization in the last four years
FILE_CARD_GIFT	Lifetime average donation from the individual in response to all card solicitations from the charitable organization
LIFETIME_CARD_PROM	Total number of card promotions sent to the individual by the charitable organization
LIFETIME_PROM	Total number of promotions (card or other) sent to the individual by the charitable organization
RECENT_AVG_CARD_GIFT_AMT	Average donation from the individual in response to a card solicitation from the charitable organization in the last four years
RECENT_RESPONSE_COUNT	Number of times the individual has responded to a promotion (card or other) from the charitable organization in the last four years
RECENT_CARD_RESPONSE_COUNT	Number of times the individual has responded to a card solicitation from the charitable organization in the last four years.
MONTHS_SINCE_LAST_PROM_RESP	Number of months since the individual has responded to a promotion by the charitable organization
LIFETIME_GIFT_AMOUNT	Total lifetime donation amount from the individual to the charitable organization
LIFETIME_GIFT_COUNT	Total number of donations from the individual to the charitable organization
LIFETIME_MAX_GIFT_AMT	Maximum donation amount from individual
LIFETIME_MIN_GIFT_AMT	Minimum donation amount from individual
LAST_GIFT_AMT	Amount donated in the individuals most recent donation
CARD_PROM_12	Number of card promotions sent to the individual by the charitable organization in the last 12 months
NUMBER_PROM_12	Number of promotions (card or other) sent to the individual by the charitable organization in the last 12 months
MONTHS_SINCE_LAST_GIFT	Number of months since the most recent donation from the individual to the charitable organization
MONTHS_SINCE_FIRST_GIFT	Number of months since the first donation from the individual to the charitable organization

IV. DELIVERABLES

Upon the project's deadline, you will be required to submit:

- A **Jupyter notebook** (or a zipfile of notebooks) featuring all the code you used throughout the project to:
 - a) Decide on your results for the objectives at hand.
 - b) Obtain your results (code that helped you make decisions, but does not directly contribute to reaching the goal, should be included, but commented).
 - c) Naming format: **DSML_GroupXX**, where **GroupXX** should be your group number.
- A **PDF of the notebook(s)** (without outputs), with a maximum of 25 A4 pages. Each extra page implies a penalization of 0.5v. Naming format: **DSML_GroupXX_pdf**, where **GroupXX** should be your group number.

V. EVALUATION

Your work will be evaluated according to the following criteria:

CRITERIA	PERCENTAGE (%)	MAXIMUM GRADE (OUT OF 20)
Project Structure & Notebook Quality	20	4
Preprocessing & EDA	25	5
Modeling & Results	40	8
Action Plan	5	1
Creativity and Self-Study	10	2

- Groups can submit all deliverables with a maximum delay of **3 days**, incurring a penalty of **1 point per day**. Beyond these three days, submissions will not be accepted.

Deadline: May 4th, 2026, 18:00

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Your grade will reflect our assessment of the quality of your work in different aspects. Please find below more details about what is taken into consideration for each topic:

- **Project Structure & Notebook Quality (4v):** Assess notebook clarity and readability, including whether the goals are clearly stated, the code contains comments, the techniques are appropriate, the insights and results are highlighted, and the implications for further steps. You should clearly outline the general approach taken and describe the various stages of the project.
- **Preprocessing & EDA (5v):** Describe the dataset and extract meaningful insights that you consider relevant to the problem. Avoid adding unnecessary visualizations or elements. This stage also includes all the steps from raw data into data ready for clustering - data cleaning, transformation, and reduction. It also entails context-related transformations of the input features, creating new features, and accompanying explanations. Here, we evaluate the quality of your implementation, its justifications, and the insights extracted.
- **Modeling & Results (8v):** Implementation and reasoning behind any clustering model used in the project. At least two perspectives should be considered; additional perspectives will be considered under the "Creativity and Self-Study" section. Additionally, you must provide an explanation of the feature selection process for each perspective. Each cluster should be statistically and visually explored and described, emphasizing the characteristics that differentiate them.
- **Action Plan (1v):** You should provide a succinct but well-oriented action plan that includes recommendations on how CSA can leverage the insights obtained from your analysis.
- **Creativity and Self-Study (2v):** If techniques not covered during the practical classes are used, a theoretical explanation of the algorithm or method should be included alongside its implementation. This topic emphasizes the application of various techniques and encourages creativity, such as alternative clustering perspectives.

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VI. PARTING NOTES

- The notebook will be the only method of evaluating your work. Ensure it is concise, focused and clear. Please don't provide long theoretical explanations of topics covered in class.
- Before submitting, run your notebook from the start one last time.
- **Your submitted notebook should include all the unneeded code you used to obtain your final solution, but it should also be commented.**
- We will run your Jupyter Notebooks if we have any doubts. So, please **make sure we can run the notebook from start to finish in one go. Notebooks that do not fulfil this condition will be penalized.**
- When determining the grade for your work, there will be a comparative component between it and the work submitted by other groups.
- **Friendly Reminders:**
 - If something is good enough to be mentioned, include it wisely.
 - Finished is better than perfect